

4.12.5.2. Reliable Message Processing of Incoming Messages

A message received from [Section 4.7.2, “Message Reception”](#) for reliability processing SHALL be processed as follows:

1. Verify the message has a legal combination of reliability flags:
 - a. If the [R Flag](#) is set:
 - i. If [Group Session Type](#) AND [C Flag](#) = 0, drop the message.
 - b. If the [A Flag](#) is set:
 - i. If [Group Session Type](#) AND [C Flag](#) = 0, drop the message.
2. Proceed to [Section 4.10.5.1, “Exchange Message Matching”](#).
3. Proceed to [Section 4.12.5.2.1, “Received acknowledgement processing”](#).

Received acknowledgement processing

1. If the [A Flag](#) is set:
 - a. Query the [retransmission table](#) for the *Acknowledgement Message Counter* contained in the received message.
 - i. If there is a match:
 - A. Remove the entry from the [retransmission table](#).
 - B. Stop the retransmission timer for that entry.
 - ii. If there is no match, it indicates that this is either a duplicate acknowledgment or the [Exchange context](#) does not exist.
2. Proceed to [Section 4.12.5.2.2, “Standalone acknowledgement processing”](#).

Standalone acknowledgement processing

1. If the [R Flag](#) is set, the received message is requesting an acknowledgement be sent back:
 - a. If the message is marked as a duplicate:
 - i. Immediately send a [standalone acknowledgment](#).
 - ii. If the Exchange is marked as an [ephemeral exchange](#) the Exchange SHALL be closed.
 - iii. Drop the message.
 - b. Otherwise, instead of sending an acknowledgement immediately upon the receipt of a reliable message from a peer, the receiver SHOULD wait for a time no longer than [MRP_STAND-ALONE_ACK_TIMEOUT](#) before sending a [standalone acknowledgment](#):
 - i. Add the message counter of the received message to the [acknowledgement table](#) to signal that an outbound acknowledgement is pending. There can be only one outstanding acknowledgement at a time on a single [Exchange](#). If a pending acknowledgement already exists for the Exchange, and it has [StandaloneAckSent](#) set to false, a [standalone acknowledgment](#) SHALL be sent immediately for that pending message counter, and the [acknowledgement table](#) entry SHALL be replaced for the new message.
 - ii. Start the acknowledgement timer for the Exchange.